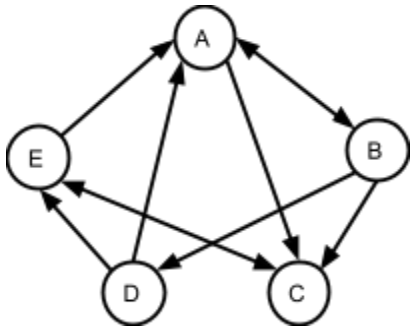
American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Intermediate Division

1. Graph Theory

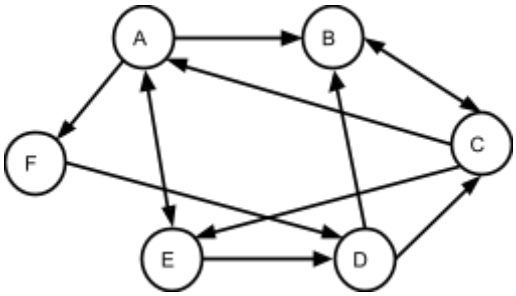
How many paths of length 2 are there in the following directed graph?



A. 15
B. 16
C. 17
D. 18
E. 19

2. Graph Theory

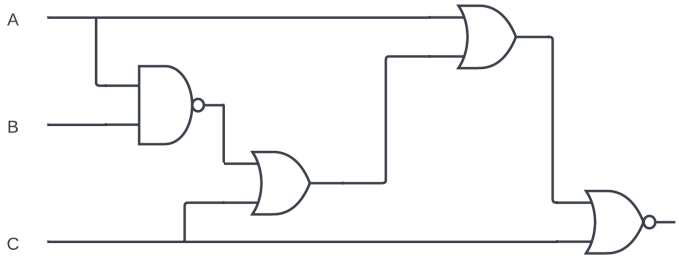
How many cycles are there that do not include A in the following directed graph?



A. 1
B. 3
C. 5
D. 8
E. 11

3. Digital Electronics

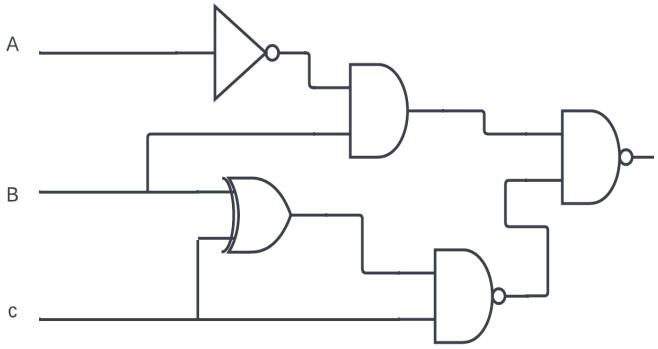
Simplify the Boolean expression represented by the digital diagram below:



A. 0
B. A
C. $\overline{B}$
D. C
E. 1

4. Digital Electronics

How many ordered triples make the following digital circuit FALSE?



A. 0
B. 2
C. 4
D. 6
E. 8

5. Assembly Language

What is the output when the following program is executed?

```
A         DC        1
C         DC        1
          LOAD      C
TOP       MULT      =100
          SUB       A
          BL        STOP
          LOAD      A
          MULT      =2
          STORE     A
          LOAD      C
          ADD       =1
          STORE     C
          BU        TOP
STOP      PRINT     C
          END
```

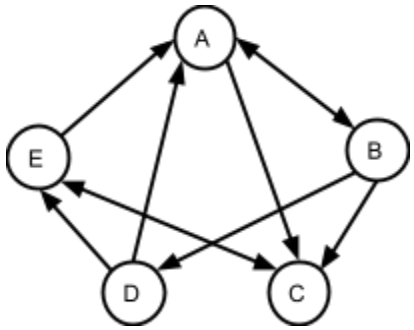
A. 10
B. 11
C. 12
D. 13
E. 14

American Computer Science League

2022-2023 ● Contest 4: Short Problems ● Intermediate Division

1. 图论

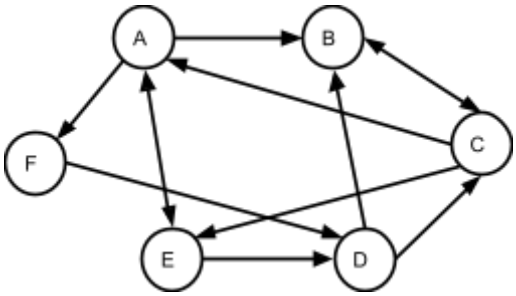
在下述有向图中，请问有多少条长度为 2 的路径？



A. 15
B. 16
C. 17
D. 18
E. 19

2. 图论

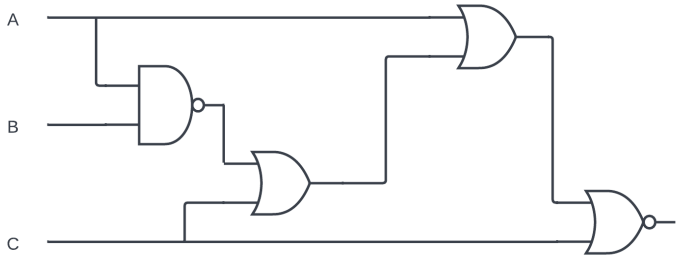
在下述有向图中，请问有多少个环不包含 A？



A. 1
B. 3
C. 5
D. 8
E. 11

3. 数字电路

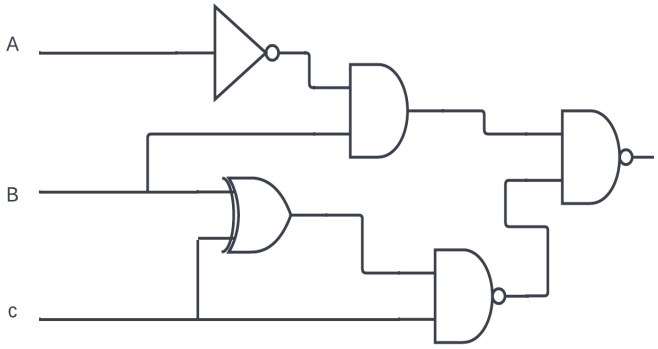
化简下述数字电路表示的布尔表达式：



A. 0
B. A
C. $\overline{B}$
D. C
E. 1

4. 数字电路

有多少个有序三元组可以使得下述数字电路为**假**？



A. 0
B. 2
C. 4
D. 6
E. 8

5. 汇编语言

执行以下程序会输出什么？

```
A         DC        1
C         DC        1
          LOAD      C
TOP       MULT      =100
          SUB       A
          BL        STOP
          LOAD      A
          MULT      =2
          STORE     A
          LOAD      C
          ADD       =1
          STORE     C
          BU        TOP
STOP      PRINT     C
          END
```

A. 10
B. 11
C. 12
D. 13
E. 14